Aryan Goel

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MS CS candidate building production data pipelines for scale, reliability, and query performance — Airflow, Snowflake, DuckDB, Kafka, Spark.

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, Core Java
Databases & Query Engines: PostgreSQL, MySQL, DuckDB (OLAP/columnar), ClickHouse, TimescaleDB, Indexing, Query Plans (EXPLAIN/EXPLAIN ANALYZE), Vectorized Execution, Storage Engines
Big Data & Pipelines: Kafka, Spark (Structured Streaming, PySpark), Dagster, dbt, Airflow (DAGs, scheduling, retries), ETL/ELT Design, Idempotent & Incremental Loading, Data Validation & Quality Checks, Hadoop
Warehousing & BI: Snowflake, Power BI, Dashboarding & Reporting
Cloud & Infra: AWS EC2, Docker, CI/CD (GitHub Actions, Jenkins Shared Libraries), GeoServer
Tools: MLflow, DVC, Git, VS Code

EXPERIENCE

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in **GeoServer** for a production WebGIS application; performed data classification with QGIS
- Redefined the rendering pipeline and resolved performance bugs to reduce load time, while shipping new data-classification features

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Engineered a cascaded CNN architecture (CED-Net) for crop/weed segmentation on RGB drone imagery, outperforming a baseline U-Net through residual blocks and attention mechanisms
- Built the full annotation-to-evaluation pipeline (LabelMe → pixel masks → training/eval on **IoU and Dice score**), benchmarking multiple architecture variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a Physics-Informed Gaussian Process Regression model to forecast flood events, embedding hydrological constraints directly into the model for improved reliability over a standard GPR baseline
- Executed uncertainty quantification via Monte Carlo and Multi-Level Monte Carlo simulations for confidence-bounded risk predictions

PROJECTS

MarketPulse: Real-Time Trading Anomaly Detector | *Python, Kafka, Spark Structured Streaming, TimescaleDB*

- Built a Kafka + Spark Structured Streaming pipeline over live Finnhub trades (stocks and BTC/ETH), bucketing into 1-minute watermarked event-time bars and persisting every trade to TimescaleDB hypertables.
- Implemented per-ticker stateful detection scoring log-return, log-volume, and trade-rate z-scores against baselines built only from completed buckets, with a 0–3 severity score and alert cooldown; detection logic is a pure, unit-tested function.
- Fixed live streaming defects (append vs. update output mode, watermark drift, Docker-network Kafka addressing); added a Gemini agent that explains flagged events.

ChainSignal: Crypto Market & On-Chain Anomaly Detection | *Python, PySpark, ClickHouse, Dagster, dbt*

- Built a batch pipeline loading 1,000 coins × 365 days of CoinGecko data and 1.64M Etherscan ERC-20 transfers into ClickHouse, paginating by block range around API limits and verifying loads against raw files (row counts, key uniqueness, 256-bit checksums).
- Orchestrated daily incremental and weekly reconcile jobs in Dagster with idempotent partition swaps, a dbt cleaning layer with 28 data-contract tests, 69 pytest tests, and GitHub Actions CI; cut transfer storage 173 → 96 MB via measured ClickHouse codec choices.
- Built no-lookahead PySpark features; compared a robust z-score, Isolation Forest, and LOF; flagged Celer Network's $1.5B volume on an $18M market cap.

Telecom Churn & Retention Intelligence | *Python, DuckDB, scikit-learn, XGBoost, GitHub Actions*

- Built a leakage-safe churn pipeline on synthetic data with a built-in Simpson's paradox (priority_support looked 6.5pp worse on raw churn but lowered it in every risk quartile once stratified), accumulating every stage into an idempotent DuckDB customer_master table.
- Enforced no-leakage by construction (split-before-fit pipelines, out-of-fold scoring, a leakage-audit test suite run in CI on every push); nested CV, isotonic calibration, and a 0.18 cost-optimal threshold cut expected cost by more than half.
- Fit S/T/X uplift models; re-ran the pipeline on real IBM Telco data (ROC-AUC 0.840), where the paradox did not replicate.

Real-Time Flight Data Engineering Pipeline | *Airflow, Snowflake, Python*

- Built a near-real-time data pipeline ingesting live flight data, orchestrated end-to-end with **Airflow DAGs** (scheduling, retries, failure alerting)
- Designed a staging-to-analytics Snowflake schema with incremental loading, reducing the manual effort required per pipeline refresh cycle

DuckDB vs. PostgreSQL: Analytical Query Performance Study — benchmarked DuckDB's columnar, vectorized engine against PostgreSQL's row-store on identical analytical queries, using EXPLAIN ANALYZE to attribute the speedup to columnar storage and vectorized execution *(DuckDB, PostgreSQL, SQL)*

ACHIEVEMENTS

- Second Runner-Up, UCLA Glitch × Google DeepMind Hackathon — built BrandSync, then developed it fully after the Google team's positive feedback
- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)